

NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2023-2024

SC4172/CE4172 – INTERNET OF THINGS: TINY MACHINE LEARNING

Apr/May 2024

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains 4 questions and comprises 2 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. All questions carry equal marks.

-
1.
 - (a) State four features that are built into Tensorflow Lite framework to facilitate development of TinyML based application.
(15 marks)
 - (b) Explain the purpose of applying data normalization.
(5 marks)
 - (c) Explain the purpose of having activation functions in a neural network model.
(5 marks)
 2.
 - (a) What are the benefits of using microcontrollers that have efficient instructions (i.e. using less clock cycles to execute an instruction) for TinyML applications?
(15 marks)
 - (b) What is the main purpose of Quantization and why is it suitable to be used for TinyML based systems?
(10 marks)

3. (a) Reference Figure Q3a. Given that NUM_GESTURES=4 and inputs_train has shape (240, 720). What is the total number of parameters (weights and bias) in the model? How many times are the model parameters updated after the entire training process? Note that the parameters are updated once during each back-propagation pass.

(15 marks)

```
# build the model and train it
model = tf.keras.Sequential()
model.add(tf.keras.layers.Dense(50, activation='relu'))
model.add(tf.keras.layers.Dense(15, activation='relu'))
model.add(tf.keras.layers.Dense(NUM_GESTURES, activation='softmax'))
model.compile(optimizer='rmsprop', loss='mse', metrics=['mae'])
history = model.fit(inputs_train, outputs_train, epochs=600, batch_size=2,
                    validation_data=(inputs_validate, outputs_validate))
```

Figure Q3a

- (b) Explain why Mel Frequency Cepstrum Coefficient (MFCC), compared to using Fast Fourier Transform (FFT) output, is a more effective frequency domain feature for speech recognition model training?

(10 marks)

4. (a) Using a color image as an example, compare the operations of Convolutional Neural Network (CNN) against the Depthwise Separable Convolution (DSC), and state the advantages of DSC.

(15 marks)

- (b) What is the main limitation of MobileNet when using it to deploy TinyML on typical microcontroller? Explain your answer.

(10 marks)

END OF PAPER